

HA 1)

$$A_T = T^{-1} A T$$

$$T^{-1} = \frac{1}{1} \begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -4 & -2 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 6 & 5 \end{pmatrix}$$

$$b_T = T^{-1} b$$

$$b_T = \begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$c_T = c T$$

$$c_T = \begin{pmatrix} 4 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 4 \end{pmatrix}$$

$$\dot{z} = T^{-1} A T z + T^{-1} b u$$

$$y = c_T z$$

$$\dot{z} = \begin{pmatrix} 0 & -1 \\ 6 & 5 \end{pmatrix} z + \begin{pmatrix} 0 \\ -1 \end{pmatrix} u$$

$$y = \begin{pmatrix} 4 & 4 \end{pmatrix} z$$

HA2) $\det(\lambda_i E - A) = 0$

$$\det \begin{pmatrix} \lambda_1 & 1 & 1 \\ 1,5 & \lambda_1 - 0,5 & 1,5 \\ -0,5 & 0,5 & \lambda_1 - 1,5 \end{pmatrix}$$

RS Vorlesung 6

$$\text{HA2)} \lambda_1 (\lambda_1 - 0,5)(\lambda_1 - 1,5) + (-0,75) + 0,75 - [0,75\lambda + 1,5(\lambda - 1,5) + (-0,5(\lambda - 0,5))] = 0$$

$$(\lambda^2 - 0,5\lambda)(\lambda - 1,5) - 0,75\lambda - 1,5\lambda + 2,25 + 0,5\lambda - 0,25 = 0$$

$$\lambda^3 - 1,5\lambda^2 - 0,5\lambda^2 + 0,75\lambda - 0,75\lambda - \lambda + 2 = 0$$

$$\lambda^3 - 2\lambda^2 - \lambda + 2 = 0$$

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

$$\lambda_3 = 2$$

$$\lambda^3 - 2\lambda^2 - \lambda + 2 : \lambda^2 - 1 = \lambda - 2$$

$$\begin{array}{r} \lambda^3 \\ 0 \end{array} \quad \begin{array}{r} -\lambda \\ 0 \end{array}$$

$$\begin{array}{r} -2\lambda^2 \\ 0 \end{array} \quad \begin{array}{r} +2 \\ 0 \end{array}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1,5 & 0,5 & 1,5 \\ -0,5 & 0,5 & -0,5 \end{pmatrix} p_1 = 0$$

$$p_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 1 & 1 \\ 1,5 & -1,5 & 1,5 \\ -0,5 & 0,5 & -2,5 \end{pmatrix} p_2 = 0$$

$$p_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} +2 & 1 & 1 \\ 1,5 & 1,5 & 1,5 \\ -0,5 & 0,5 & 0,5 \end{pmatrix} p_3 = 0$$

$$\cancel{p_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}} \quad p_3 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

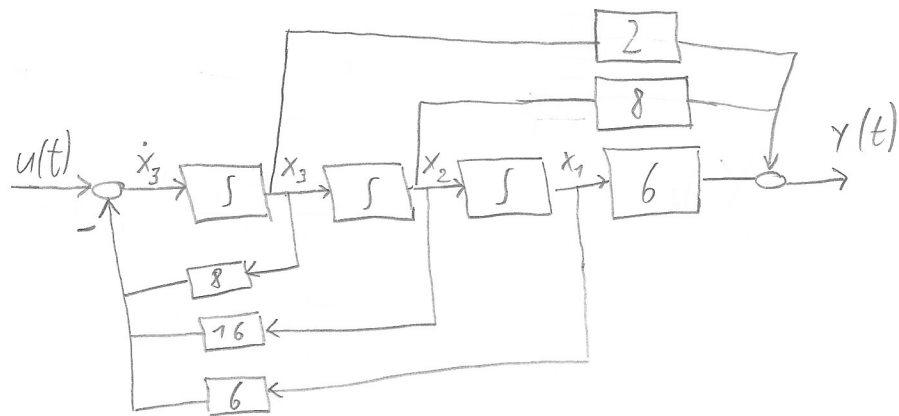
$$T = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix}$$

RS Vorlesung 6

HA 3)

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -16 & -8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u$$

$$y = \begin{pmatrix} 6 & 8 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$



HA4)

$$y = \begin{pmatrix} 0 & R \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\dot{X} = \begin{pmatrix} 0 & -\frac{1}{C} \\ \frac{1}{L} & \frac{R}{L} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} \frac{1}{C} \\ 0 \end{pmatrix} u$$